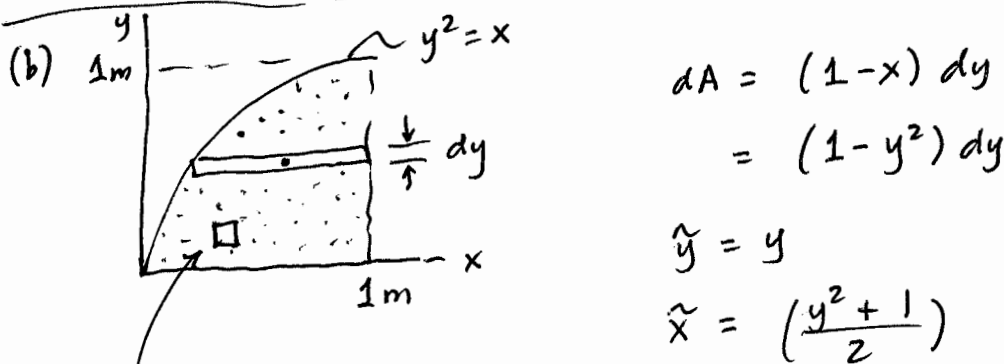
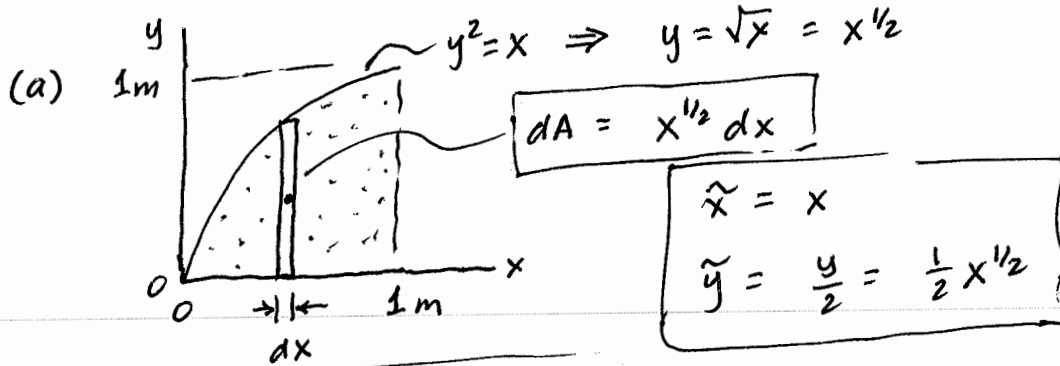
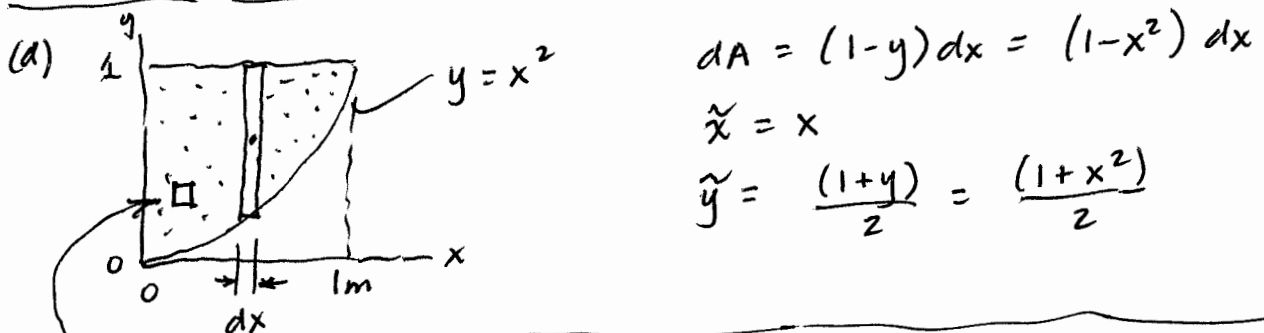
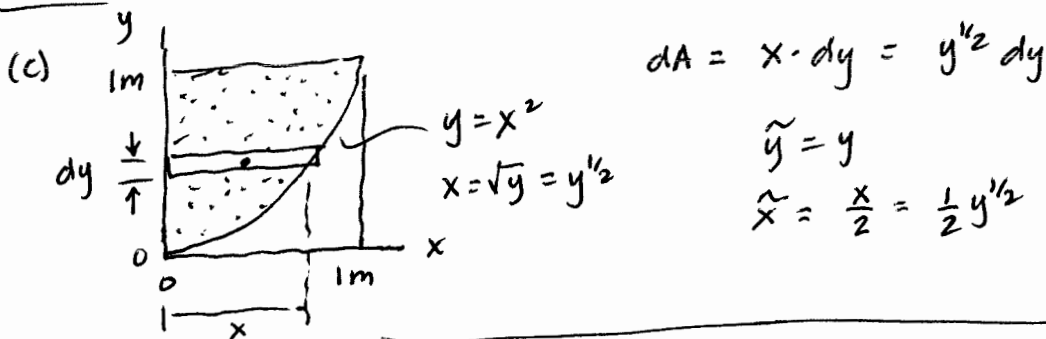


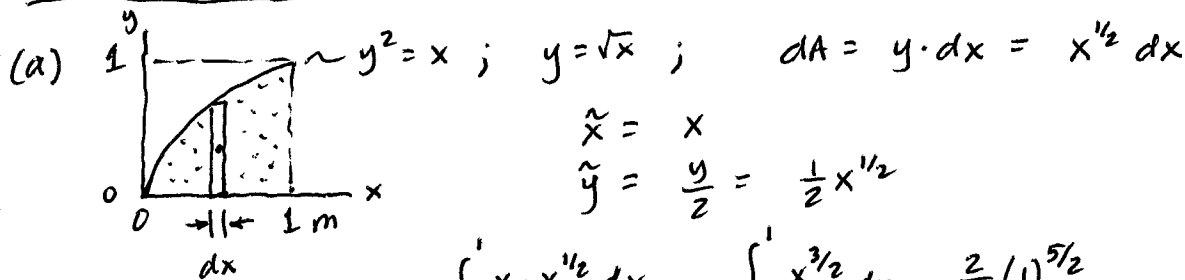
For each problem, specify $\tilde{x}$, $\tilde{y}$, dA :



(bb) $dA = dx dy$, $\tilde{x} = x$, $\tilde{y} = y$
 limits: y from 0 to $\sqrt{x}$; x from 0 to 1



(dd) $dA = dx dy$, $\tilde{x} = x$, $\tilde{y} = y$
 limits: y from x^2 to 1; x from 0 to 1.



$$\tilde{x} = x$$

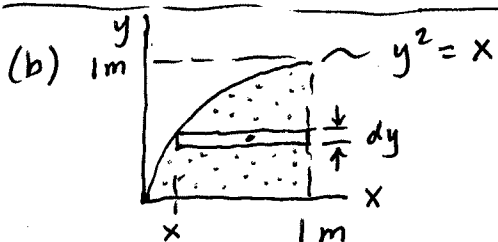
$$\tilde{y} = \frac{y}{2} = \frac{1}{2} x^{1/2}$$

$$\bar{x} = \frac{\int_0^1 x \cdot x^{1/2} dx}{\int_0^1 x^{1/2} dx} = \frac{\int_0^1 x^{3/2} dx}{\frac{2}{3}(1)^{3/2}} = \frac{\frac{2}{5}(1)^{5/2}}{2/3} = \frac{2}{5} \cdot \frac{3}{2} = \frac{3}{5}$$

$$\boxed{\bar{x} = 0.60} \text{ m}$$

$$\bar{y} = \frac{\int_0^1 (\frac{1}{2} x^{1/2}) (x^{1/2} dx)}{2/3} = \frac{\int_0^1 \frac{1}{2} x dx}{2/3} = \frac{\frac{1}{4}(1)^2}{2/3} = \frac{1}{4} \cdot \frac{3}{2} = \frac{3}{8}$$

$$\boxed{\bar{y} = \frac{3}{8} = 0.375} \text{ m}$$



$$dA = (1 - x) dy = (1 - y^2) dy$$

$$\tilde{y} = y$$

$$\tilde{x} = \frac{1}{2}(1 + x) = \frac{1}{2}(y^2 + 1) = \frac{5}{10} - \frac{1}{10}$$

$$\bar{x} = \frac{\int_0^1 \frac{1}{2}(y^2 + 1)(1 - y^2) dy}{\int_0^1 (1 - y^2) dy} = \frac{\int_0^1 \frac{1}{2}(1 - y^4) dy}{(y - \frac{1}{3}y^3)'_0} = \frac{(\frac{1}{2} - \frac{1}{10}y^5)'_0}{1 - 1/3} = \frac{4/10}{2/5} = 2/5$$

$$\bar{x} = \frac{2/5}{2/3} = \frac{2}{5} \cdot \frac{3}{2} = \boxed{\frac{3}{5} = 0.60 \text{ m} = \bar{x}}$$

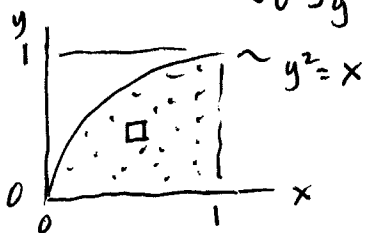
$$\bar{y} = \frac{\int_0^1 y(1 - y^2) dy}{2/3} = \frac{\int_0^1 y - y^3 dy}{2/3} = \frac{(\frac{1}{2}y^2 - \frac{1}{4}y^4)'_0}{2/3} = \frac{\frac{1}{2} - 1/4}{2/3} = \frac{1/4}{2/3}$$

$$\bar{y} = \frac{1}{4} \cdot \frac{3}{2} = \boxed{\frac{3}{8} = 0.375 \text{ m} = \bar{y}}$$

(bb) $\text{Area} = \int_0^1 \int_0^{\sqrt{x}} \underbrace{dy dx}_{=dA} = \int_0^1 x^{1/2} dx = \frac{2}{3} x^{3/2} \Big|_0^1 = \boxed{\frac{2}{3} \text{ m}^2 = A}$

or $\text{Area} = \int_0^1 \int_{y^2}^1 \underbrace{dx dy}_{=dA} = \int_0^1 (1 - y^2) dy = (y - \frac{1}{3}y^3)'_0 = 1 - 1/3$

$$\boxed{A = \frac{2}{3} \text{ m}^2}$$



$$dA = dx dy$$

or $dy dx$

(bb) $\bar{x}$ numerator ($dA = dy dx$ w/ approp. limits)

$$= \int_0^1 \int_0^{\sqrt{x}} x dy dx = \int_0^1 x(x^{1/2}) dx = \int_0^1 x^{3/2} dx = \frac{2}{5} x^{5/2} \Big|_0^1 = \frac{2}{5}$$

$$\bar{x} = \frac{2/5}{2/3} = \frac{2}{5} \cdot \frac{3}{2} = \boxed{\frac{3}{5} = 0.6 \text{ m} = \bar{x}}$$

or $\bar{x}$ numerator ($dA = dx dy$ w/ approp. limits)

$$= \int_0^1 \int_{y^2}^1 x dx dy = \int_0^1 \frac{1}{2} x^2 \Big|_{y^2}^1 dy = \int_0^1 \frac{1}{2} (1 - y^4) dy$$

$$= \left[\frac{1}{2} y - \frac{1}{10} y^5 \right]_0^1 = \frac{1}{2} - \frac{1}{10} = \frac{5}{10} - \frac{1}{10} = \frac{4}{10} = \frac{2}{5}$$

$$\bar{x} = \frac{2/5}{2/3} = \frac{2}{5} \cdot \frac{3}{2} = \boxed{\frac{3}{5} = 0.6 \text{ m} = \bar{x}}$$

(bb) $\bar{y}$ numerator ($dA = dy dx$ w/ approp. limits)

$$= \int_0^1 \int_0^{\sqrt{x}} y dy dx = \int_0^1 \frac{1}{2} y^2 \Big|_0^{\sqrt{x}} dx = \int_0^1 \frac{1}{2} x dx = \frac{1}{4} x^2 \Big|_0^1 = \frac{1}{4}$$

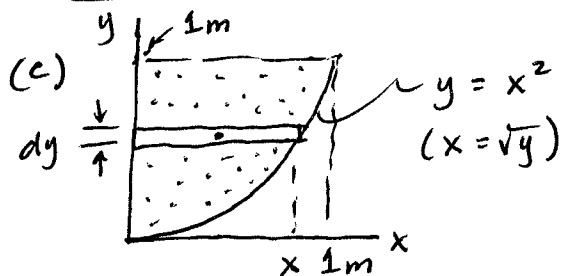
$$\bar{y} = \frac{1/4}{2/3} = \frac{1}{4} \cdot \frac{3}{2} = \boxed{\frac{3}{8} = 0.375 \text{ m} = \bar{y}}$$

or $\bar{y}$ numerator ($dA = dx dy$ w/ approp. limits)

$$= \int_0^1 \int_{y^2}^1 y dx dy = \int_0^1 y(1 - y^2) dy = \int_0^1 (y - y^3) dy$$

$$= \left(\frac{1}{2} y^2 - \frac{1}{4} y^4 \right) \Big|_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$\bar{y} = \frac{1/4}{2/3} = \frac{1}{4} \cdot \frac{3}{2} = \boxed{\frac{3}{8} = 0.375 \text{ m} = \bar{y}}$$



$$dA = x \cdot dy = \sqrt{y} \cdot dy$$

$$\tilde{x} = \frac{x}{2} = \frac{1}{2}\sqrt{y}$$

$$\tilde{y} = y$$

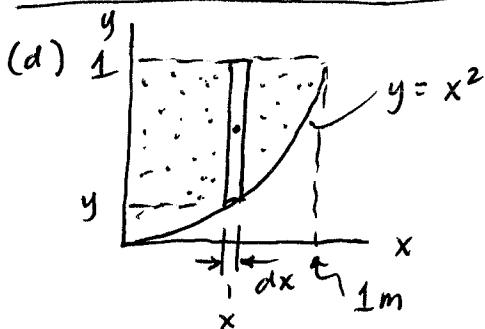
$$\text{Area} = \int_0^1 y^{1/2} dy = \frac{2}{3} y^{3/2} \Big|_0^1 = \boxed{\frac{2}{3} \text{ m}^2 = A}$$

$$\bar{x} \text{ numerator} = \int_0^1 \left(\frac{1}{2} y^{1/2} \right) (y^{1/2} dy) = \int_0^1 \frac{1}{2} y dy = \frac{1}{4} y^2 \Big|_0^1 = \frac{1}{4}$$

$$\bar{x} = \frac{1/4}{2/3} = \frac{1}{4} \cdot \frac{3}{2} = \boxed{\frac{3}{8} = 0.375 \text{ m} = \bar{x}}$$

$$\bar{y} \text{ numerator} = \int_0^1 y (y^{1/2} dy) = \int_0^1 y^{3/2} dy = \frac{2}{5} y^{5/2} \Big|_0^1 = \frac{2}{5}$$

$$\bar{y} = \frac{2/5}{2/3} = \frac{2}{5} \cdot \frac{3}{2} = \boxed{\frac{3}{5} = 0.6 \text{ m} = \bar{y}}$$



$$dA = (1 - y) dx = (1 - x^2) dx$$

$$\tilde{x} = x$$

$$\tilde{y} = \left(\frac{1 - y}{2} \right) = \frac{1}{2}(1 + x^2)$$

$$\text{Area} = \int_0^1 (1 - x^2) dx = \left(x - \frac{1}{3} x^3 \right) \Big|_0^1 = 1 - \frac{1}{3} = \boxed{\frac{2}{3} \text{ m}^2 = A}$$

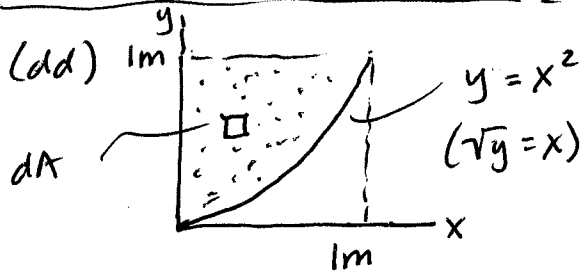
$$\begin{aligned} \bar{x} \text{ Numerator} &= \int_0^1 x (1 - x^2) dx = \int_0^1 (x - x^3) dx = \left(\frac{1}{2} x^2 - \frac{1}{4} x^4 \right) \Big|_0^1 \\ &= \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \end{aligned}$$

$$\bar{x} = \frac{1/4}{2/3} = \frac{1}{4} \cdot \frac{3}{2} = \boxed{\frac{3}{8} = 0.375 \text{ m} = \bar{x}}$$

(d) (cont'd)

$$\begin{aligned}\bar{y} \text{ Numerator} &= \int_0^1 \underbrace{\frac{1}{2}(1+x^2)}_{\tilde{y}} \underbrace{(1-x^2)}_{dA} dx \\ &= \int_0^1 \frac{1}{2}(1-x^4) dx = \left(\frac{1}{2}x - \frac{1}{10}x^5 \right) \Big|_0^1 = \frac{1}{2} - \frac{1}{10} \\ &= \frac{5}{10} - \frac{1}{10} = \frac{4}{10} = \frac{2}{5}\end{aligned}$$

$$\bar{y} = \frac{2/5}{2/3} = \frac{2}{5} \cdot \frac{3}{2} = \boxed{\frac{3}{5} = 0.6 \text{ m} = \bar{y}}$$



$$dA = dy dx$$

$$\text{or } dx dy$$

$$\tilde{x} = x ; \quad \tilde{y} = y$$

$$\text{Area} = \int_0^1 \int_{x^2}^1 dy dx = \int_0^1 (1-x^2) dx = \left(x - \frac{1}{3}x^3 \right) \Big|_0^1$$

$$A = 1 - \frac{1}{3} = \boxed{\frac{2}{3} \text{ m}^2 = A}$$

$$\text{or } A = \int_0^1 \int_0^{\sqrt{y}} dx dy = \int_0^1 y^{1/2} dy = \left. \frac{2}{3} y^{3/2} \right|_0^1 = \boxed{\frac{2}{3} \text{ m}^2 = A}$$

(dd) $\bar{x}$ numerator (Using $dA = dy dx$ + approp. limits)

$$\begin{aligned}&= \int_0^1 \int_{x^2}^1 x dy dx = \int_0^1 x(1-x^2) dx = \int_0^1 (x-x^3) dx \\ &= \left(\frac{1}{2}x^2 - \frac{1}{4}x^4 \right) \Big|_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}\end{aligned}$$

$$\bar{x} = \frac{1/4}{2/3} = \frac{1}{4} \cdot \frac{3}{2} = \boxed{\frac{3}{8} = 0.375 \text{ m} = \bar{x}}$$

(dd) (cont'd)

 $\bar{x}$ numerator (using $dA = dx dy + \text{approp. limits}$)

$$= \int_0^1 \int_0^{\sqrt{y}} x dx dy = \int_0^1 \frac{1}{2} x^2 \Big|_0^{\sqrt{y}} dy = \int_0^1 \frac{1}{2} y dy = \frac{1}{4} y^2 \Big|_0^1$$

$$\bar{x} = \frac{1/4}{2/3} = \frac{1}{4} \cdot \frac{3}{2} = \boxed{\frac{3}{8} = 0.375 \text{ m} = \bar{x}} = \frac{1}{4}$$

 $\bar{y}$ numerator (using $dA = dy dx + \text{approp limits}$)

$$= \int_0^1 \int_{x^2}^1 y dy dx = \int_0^1 \frac{1}{2} y^2 \Big|_{x^2}^1 dx = \int_0^1 \frac{1}{2} (1 - x^4) dx$$

$$= \left(\frac{1}{2} (x) - \frac{1}{10} x^5 \right) \Big|_0^1 = \frac{1}{2} - \frac{1}{10} = \frac{5}{10} - \frac{1}{10} = \frac{4}{10} = \frac{2}{5}$$

$$\bar{y} = \frac{2/5}{2/3} = \boxed{\frac{3}{5} = 0.6 \text{ m} = \bar{y}}$$

 $\bar{y}$ numerator (using $dA = dx dy + \text{approp limits}$)

$$= \int_0^1 \int_0^{\sqrt{y}} y dx dy = \int_0^1 y (y^{1/2}) dy = \int_0^1 y^{3/2} dy = \frac{2}{5} y^{5/2} \Big|_0^1 = \frac{2}{5}$$

$$\bar{y} = \frac{2/5}{2/3} = \boxed{\frac{3}{5} = 0.6 \text{ m} = \bar{y}}$$